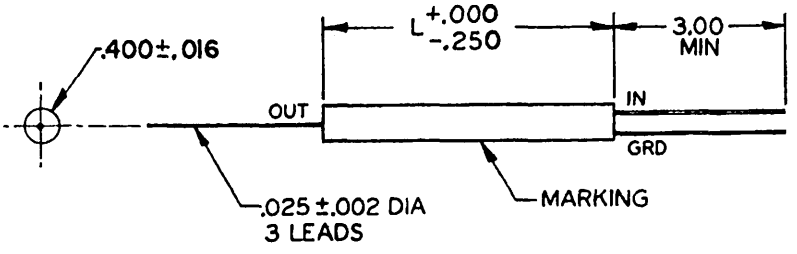
MILITARY SPECIFICATION SHEET

DELAY LINE, PULSE, ELECTROMAGNETIC, FIXED

STYLE RA

The complete requirements for procuring the delay line described herein shall consist of this document and the latest issue of Specification MIL-D-23859.

| INCHES | MM |
|---|---|
| .002 | .05 |
| .016 | .41 |
| .025 | .64 |
| .250 | 6.35 |
| .400 | 10.16 |
| 3.00 | 76.20 |




NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. General configuration.

Type designation, terminating impedance, delay time, figure of merit, attenuation, dimensions and rise time.

| Part number | Type designation | Nominal terminating impedance (ohms) | Delay time (usec) | Figure of merit | Attenuation percent | Dimensions L Inches | Dimensions L MM | Max. rise time usec |
|---|---|---|---|---|---|---|---|---|
| M23859/13-001 | DL501RA2006B | 500 | .20 | B | 10 | 2.531 | (64.29) | .074 |
| M23859/13-002 | DL501RA2206B | 500 | .22 | B | 10 | 2.812 | (71.42) | .081 |
| M23859/13-003 | DL501RA2406B | 500 | .24 | B | 11 | 2.938 | (74.63) | .089 |
| M23859/13-004 | DL501RA2706C | 500 | .27 | C | 12 | 3.219 | (81.76) | .048 |
| M23859/13-005 | DL501RA3006C | 500 | .30 | C | 12 | 3.406 | (88.51) | .053 |
| M23859/13-006 | DL501RA3306C | 500 | .33 | C | 13 | 3.750 | (95.25) | .059 |
| M23859/13-007 | DL501RA3606D | 500 | .36 | D | 14 | 3.969 | (100.81) | .053 |
| M23859/13-008 | DL501RA3906D | 500 | .39 | D | 14 | 4.219 | (107.16) | .057 |
| M23859/13-009 | DL501RA4306D | 500 | .43 | D | 15 | 4.594 | (116.69) | .063 |
| M23859/13-010 | DL501RA4706D | 500 | .47 | D | 16 | 4.812 | (122.22) | .069 |
| M23859/13-011 | DL501RA5106E | 500 | .51 | E | 16 | 5.219 | (132.56) | .062 |
| M23859/13-012 | DL501RA5606E | 500 | .56 | E | 17 | 5.625 | (142.88) | .068 |
| M23859/13-013 | DL501RA6206E | 500 | .62 | E | 18 | 6.219 | (157.96) | .075 |
| M23859/13-014 | DL501RA6806E | 500 | .68 | E | 19 | 6.719 | (170.66) | .083 |
| M23859/13-015 | DL501RA7506F | 500 | .75 | F | 20 | 7.344 | (186.58) | .075 |
| M23859/13-016 | DL501RA8206F | 500 | .82 | F | 22 | 7.969 | (202.41) | .082 |
| M23859/13-017 | DL501RA9106F | 500 | .91 | F | 24 | 8.750 | (222.25) | .091 |
| M23859/13-018 | DL501RA1005F | 500 | 1.00 | F | 25 | 9.531 | (242.09) | .100 |
| M23859/13-019 | DL501RA1105G | 500 | 1.10 | G | 27 | 10.406 | (264.31) | .092 |
| M23859/13-020 | DL102RA2006C | 1000 | .20 | C | 10 | 2.406 | (63.11) | .036 |



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Type designation, terminating impedance, delay time, figure of merit, attenuation, dimensions and rise time. (Cont'd.)

| Part number | Type designation | Nominal terminating impedance (ohms) | Delay time (usec) | Figure of merit | Attenuation percent | Dimensions L | | Max. rise time usec |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Inches | MM | |
| M23859/13-021 | DL102RA2206C | 1000 | .22 | C | 10 | 2.703 | (68.66) | .039 |
| M23859/13-022 | DL102RA2406C | 1000 | .24 | C | 11 | 2.859 | (72.62) | .043 |
| M23859/13-023 | DL102RA2706C | 1000 | .27 | C | 12 | 3.125 | (79.38) | .048 |
| M23859/13-024 | DL102RA3006C | 1000 | .30 | C | 12 | 3.406 | (88.51) | .053 |
| M23859/13-025 | DL102RA3306D | 1000 | .33 | D | 13 | 3.609 | (91.67) | .048 |
| M23859/13-026 | DL102RA3606D | 1000 | .36 | D | 14 | 3.859 | (98.02) | .053 |
| M23859/13-027 | DL102RA3906D | 1000 | .39 | D | 14 | 4.125 | (104.78) | .057 |
| M23859/13-028 | DL102RA4306E | 1000 | .43 | E | 15 | 4.453 | (113.11) | .052 |
| M23859/13-029 | DL102RA4706E | 1000 | .47 | E | 16 | 4.796 | (121.82) | .057 |
| M23859/13-030 | DL102RA5106E | 1000 | .51 | E | 16 | 5.109 | (129.77) | .062 |
| M23859/13-031 | DL102RA5606E | 1000 | .56 | E | 17 | 5.531 | (140.49) | .068 |
| M23859/13-032 | DL102RA6206F | 1000 | .62 | F | 18 | 6.047 | (153.59) | .062 |
| M23859/13-033 | DL102RA6806F | 1000 | .68 | F | 19 | 6.546 | (166.27) | .068 |
| M23859/13-034 | DL102RA7506F | 1000 | .75 | F | 20 | 7.125 | (180.98) | .075 |
| M23859/13-035 | DL102RA8206F | 1000 | .82 | F | 22 | 7.703 | (195.66) | .082 |
| M23859/13-036 | DL102RA9106G | 1000 | .91 | G | 24 | 8.500 | (215.90) | .076 |
| M23859/13-037 | DL102RA1005G | 1000 | 1.00 | G | 25 | 9.282 | (235.76) | .083 |
| M23859/13-038 | DL102RA1105G | 1000 | 1.10 | G | 27 | 10.093 | (256.36) | .092 |
| M23859/13-039 | DL152RA2006B | 1500 | .20 | B | 10 | 2.031 | (51.59) | .074 |
| M23859/13-040 | DL152RA2206B | 1500 | .22 | B | 10 | 2.125 | (53.98) | .082 |
| M23859/13-041 | DL152RA2406B | 1500 | .24 | B | 11 | 2.219 | (56.36) | .086 |
| M23859/13-042 | DL152RA2706B | 1500 | .27 | B | 12 | 2.375 | (60.33) | .100 |
| M23859/13-043 | DL152RA3006C | 1500 | .30 | C | 13 | 2.531 | (64.29) | .054 |
| M23859/13-044 | DL152RA3306C | 1500 | .33 | C | 13 | 2.781 | (70.64) | .059 |
| M23859/13-045 | DL152RA3606C | 1500 | .36 | C | 14 | 3.125 | (79.38) | .064 |
| M23859/13-046 | DL152RA3906D | 1500 | .39 | D | 15 | 3.156 | (80.16) | .057 |
| M23859/13-047 | DL152RA4306D | 1500 | .43 | D | 16 | 3.375 | (85.73) | .063 |
| M23859/13-048 | DL152RA4706D | 1500 | .47 | D | 17 | 3.406 | (86.51) | .069 |
| M23859/13-049 | DL152RA5106E | 1500 | .51 | E | 18 | 3.780 | (96.01) | .062 |
| M23859/13-050 | DL152RA5606E | 1500 | .56 | E | 19 | 4.031 | (102.39) | .068 |
| M23859/13-051 | DL152RA6206E | 1500 | .62 | E | 20 | 4.406 | (111.91) | .076 |
| M23859/13-052 | DL152RA6806E | 1500 | .68 | E | 22 | 4.750 | (120.65) | .083 |
| M23859/13-053 | DL152RA7506E | 1500 | .75 | E | 23 | 5.188 | (131.78) | .092 |
| M23859/13-054 | DL152RA8206F | 1500 | .82 | F | 25 | 5.406 | (137.31) | .082 |
| M23859/13-055 | DL152RA9106F | 1500 | .91 | F | 27 | 6.156 | (156.36) | .091 |
| M23859/13-056 | DL152RA1005F | 1500 | 1.00 | F | 30 | 6.656 | (169.06) | .100 |
| M23859/13-057 | DL152RA1105G | 1500 | 1.10 | G | 32 | 7.281 | (184.94) | .092 |
| M23859/13-058 | DL152RA1205G | 1500 | 1.20 | G | 34 | 7.781 | (197.64) | .100 |
| M23859/13-059 | DL152RA1305G | 1500 | 1.3 | G | 36 | 8.406 | (213.51) | .108 |
| M23859/13-060 | DL152RA1505G | 1500 | 1.5 | G | 41 | 9.531 | (242.09) | .125 |

ENGINEERING DATA:

Case material: Metallic (nonferrous).

Operating temperature range: -65°C to +95°C.

Input pulse:
Peak pulse voltage: 150 volts maximum.
Pulse width: Shall be greater than 3 times the output rise time and less than 1.5 times the delay time.
Pulse rise time: 15 to 20 percent of delay line rise time.

Nominal terminating impedance: See table.

Style: RA.

Figure of merit: See table.



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REQUIREMENTS:

Design and construction: See figure 1.

Enclosure:
Type: Tubular.
Type of seal: Hermetic.

Leads: Solid tinned copper wire (output wire axially located).

Delay time: See table.

Rise time: See table.

Attenuation, percent maximum: See table.

Distortion: ±10 percent at maximum delay.

Dielectric withstanding voltage: Method 301 of MIL-STD-202.

Delay time variation with temperature: ±2 percent maximum of nominal delay from -65 to 95 C.

Insulation resistance: Method 302, test condition A, MIL-STD-202. 1,000 megohms minimum.

Thermal age stability: 3 percent maximum for delay; Rise time, attenuation and distortion shall still meet the requirements of this specification sheet.

Sealing: As specified in 4.7.3.1.1.

Terminal strength: Method 211, test condition D, MIL-STD-202.

Salt atmosphere: Method 101, test condition A, MIL-STD-202.
Salt solution - 5 percent.
Measurements - Not applicable.

Pressure: Method 105, test condition C, MIL-STD-202..
Test voltage - 250 vdc applied between IN and GRD with GRD connected to case.
Mounting - Normal.
Measurements - After exposure, sealing test.

Vibration: Method 204, test condition C, MIL-STD-202.
Mounting - Rigidly mounted at two points by clamps or clips, not closer than 0.120 from each end.
Measurements - Not applicable.

Shock: Method 205, test condition A, MIL-STD-202.
Mounting - Rigidly mounted at two points by clamps or clips, not closer than 0.120 from each end.
Measurements - Not applicable.

Thermal shock: Method 107, test condition A, MIL-STD-202, except upper temperature limit is 95°C, and low temperature shall be -65°C.
Measurements - Not applicable.

Immersion: Method 104, test condition A, MIL-STD-202.
Measurements - Insulation resistance.

Moisture resistance: Method 106 of MIL-STD-202.
Loading voltage - Not applicable.
Polarizing voltage - 100 volts dc.
Measurements - Insulation resistance and dielectric withstanding voltage.

Flammability: Not applicable.

Life: Method 108, test condition D, MIL-STD-202.
Test temperature - 95° ±2°C.
Test voltage - 150 vdc applied between IN and GRD with GRD connected to case.
Measurements - Dielectric withstanding voltage, insulation resistance, delay time variation with temperature and thermal age stability.



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Part number: Consists of the basic number of this specification sheet and a dash number taken from table.

Example: M23859/13-001

Basic document number

Dash number from table

Custodians:
Army - EL
Navy - SH
Air Force - 80

Review activities:
Army - MU, EL
Navy - SH, AS, OS
Air Force - 80, 17, 85

User activities:
Army - MI
Navy - MC
Air Force - None

Preparing activity:
Navy - SH

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Code "C"



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